

Vectors, Geometry & Similarity

Mathematics Foundations — Session 1

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Quiz

[Open the quiz](#)

Today's Goal

- Rebuild vector intuition in $\mathbb{R}^n$: addition, scalar multiplication, 2D/3D
 - Measure the alignment between two vectors
 - Read the dot product as a measure of that alignment
 - Use cosine similarity to compare real data
- Deliverable: rank the most similar items in a real dataset using cosine similarity

Pen and paper

This class is pen-and-paper oriented: we build the intuition by hand, on the board and on your notebook, before any code.

Plan

1. Vectors in $\mathbb{R}^n$
2. Dot Product and Euclidean Norm
3. First Set of Exercises
4. Distance and Cosine Similarity
5. Python Example (NumPy)
6. Link with ML
7. Final Exercises
8. Wrap-Up

Vectors in $\mathbb{R}^n$

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Vectors: What Are They?

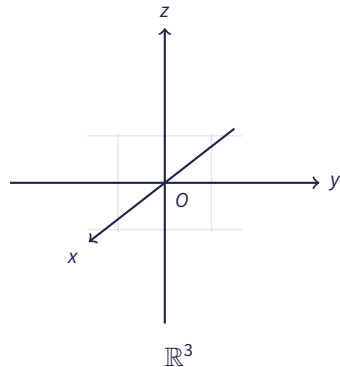
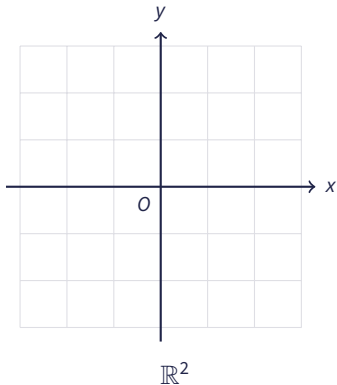
Vectors are mathematical objects

A vector is a mathematical object, defined on its own terms. Machine learning uses vectors to store information. We will therefore review the properties of vectors as the mathematical objects they are.

Vector Size

What Can We Store in a Vector?

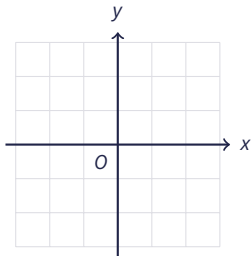
Vectors: Geometrical Meaning



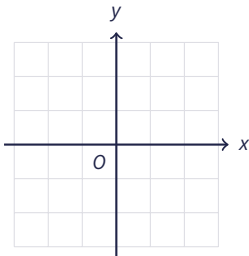
Vector Properties

- Vector equalities
- Addition
- Scalar multiplication

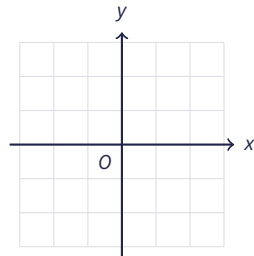
Vector Properties: Geometrical Meaning



Vector equalities

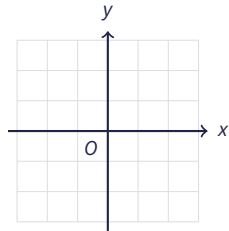


Addition



Scalar multiplication

Vector Decomposition



Can we always do this?

Vector Independence

Question

Are these vectors linearly independent?

Notion of Basis

Dot Product and Euclidean Norm

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- 2. Dot Product and Euclidean Norm**
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Dot Product Operation (1/2)

Dot Product Operation (2/2)

Dot Product Properties

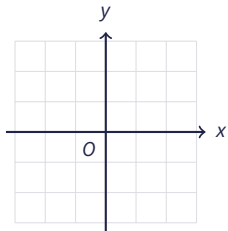
Symmetry

Linearity

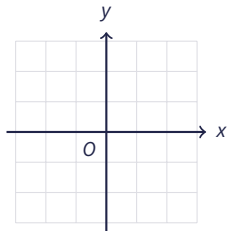
**Positive
definiteness**

Euclidean Norm

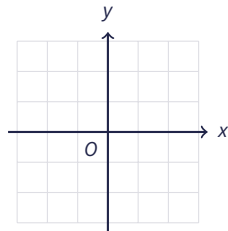
Orthogonal and Orthonormal Basis



Basis



Orthogonal basis



Orthonormal basis

Norms (General Case)

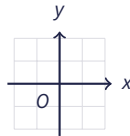
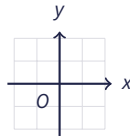
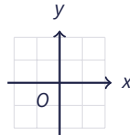
Definition: norm

A norm on $\mathbb{R}^n$ is a function $\| \cdot \| : \mathbb{R}^n \rightarrow \mathbb{R}$ such that, for all $x, y \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$:

1. $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$
2. $\|\lambda x\| = |\lambda| \|x\|$
3. $\|x + y\| \leq \|x\| + \|y\|$

Norms (General Case)

L^1
 L^2
 L^∞



Normalization

First Set of Exercises

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Exercises (1/2)

Exercise 1

Exercises: Norm and Distance (2/2)

Exercise 2 — Norm and distance

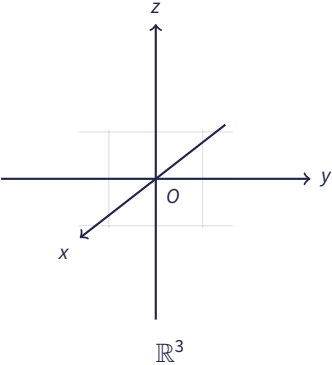
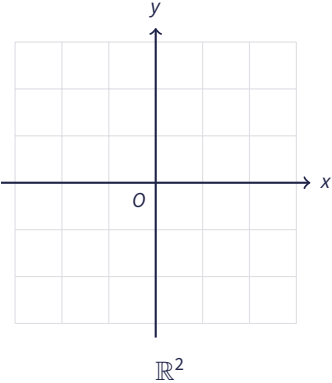
Distance and Cosine Similarity

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Link with ML

Distance

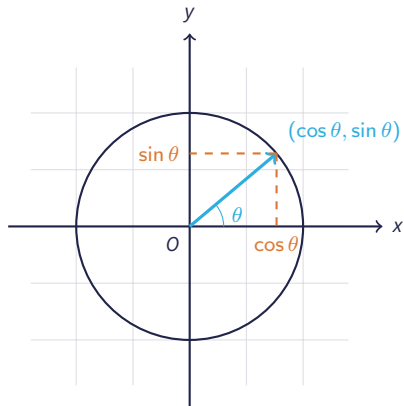
Distance in Geometry



Angle (1/5)

Angle (2/5)

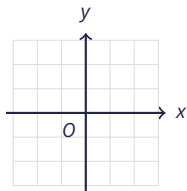
Angle (3/5)



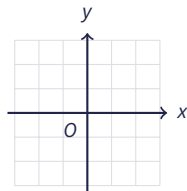
Angle (4/5)

Angle (5/5)

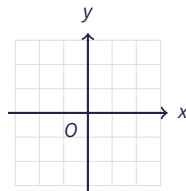
Angle Examples in 2D and 3D



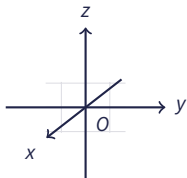
2D: $(1, 0)$ and $(1, 1)$



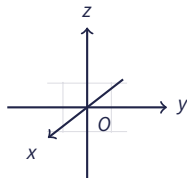
2D: $(1, 0)$ and $(0, 2)$



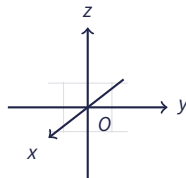
2D: $(1, 1)$ and $(-1, -1)$



3D: $(1, 0, 0)$ and $(0, 1, 0)$



3D: $(1, 1, 0)$ and $(1, 1, 1)$



3D: $(1, 0, 0)$ and $(-1, 0, 0)$

Cosine Similarity

Link Between the Dot Product, Norm and Distance

Cosine and Similarity (1/2)

Exercise

Cosine and Similarity (2/2)

Exercise

Python Example (NumPy)

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Python Example Using NumPy

Link with ML

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Link with ML

Link with LLMs

Neural Networks

Intro to Matrices

Final Exercises

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Recap Exercises (1/2)

Recap exercise 1

Recap Exercises (2/2)

Recap exercise 2

Wrap-Up

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Recap

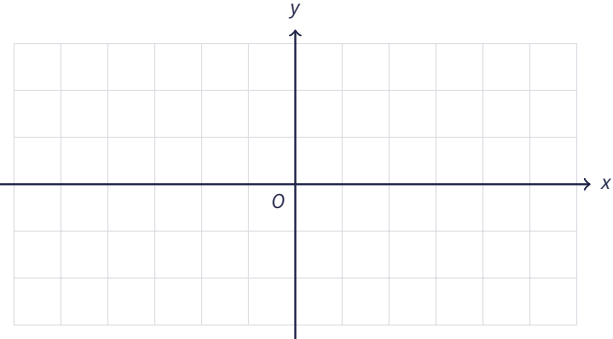
To Do for Next Session

Class Feedback

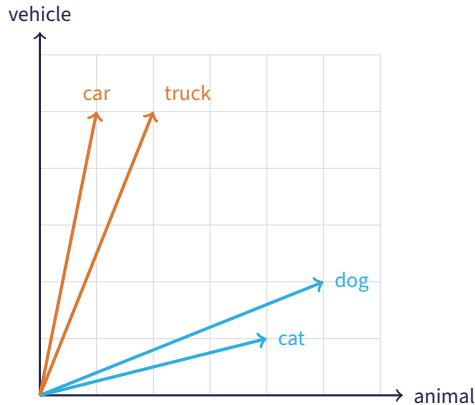
Appendix

9. Appendix

Coordinate Grid



Backup: Word Embeddings (Toy Example)



Each word $\rightarrow$ a vector in $\mathbb{R}^2$ (axes: “animal-ness”, “vehicle-ness”).

word	animal	vehicle
cat	4	1
dog	5	2
car	1	5
truck	2	5

Real embeddings (word2vec, GloVe, LLMs):
hundreds to thousands of dimensions,
learned from text, not hand-designed.

Backup: Similarity Between Words

$$\cos(\text{cat}, \text{dog}) = \frac{4 \cdot 5 + 1 \cdot 2}{\sqrt{17} \sqrt{29}} \approx 0.99$$

$$\cos(\text{cat}, \text{car}) = \frac{4 \cdot 1 + 1 \cdot 5}{\sqrt{17} \sqrt{26}} \approx 0.43$$

	cat	dog	car	truck
cat	1	0.99	0.43	0.59
dog	0.99	1	0.55	0.69
car	0.43	0.55	1	0.98
truck	0.59	0.69	0.98	1

Reading

cat/dog and car/truck are nearly aligned (cosine close to 1): similar words point in similar directions.

Backup: Word Analogies (Toy Example)

Toy vectors in $\mathbb{R}^4$, axes: royalty, male, female, animal.

word	royalty	male	female	animal
king	9	9	1	0
queen	9	1	9	0
man	1	9	1	0
woman	1	1	9	0
cat	0	4	4	9

$$\text{king} - \text{man} + \text{woman} = (9, 1, 9, 0) = \text{queen}$$

Cosine with the result

queen 1.00 woman 0.78 king 0.61 cat 0.30 man 0.23